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Hiring Committee
Digonex Technologies, Inc.
9102 N. Meridian Street, Suite 300
Indianapolis, IN 46260

Dear Hiring Committee,

I am writing to express my interest in an Economist or Research Scientist position at Digonex. I am a Ph.D. candidate in the Department of Economics at Johns Hopkins University (advisors: Christopher D. Carroll, Nicholas W. Papageorge, and Stelios Fourakis), and I expect to receive my degree in May 2026. My primary specializations are macroeconomic modeling, wealth and income dynamics, and computational economics.

My research agenda focuses on the ability of macroeconomic models to generate empirically realistic distributions of wealth and income over the life cycle. A key driver of that work is modeling heterogeneity in the rate of return on assets, and understanding how return dispersion contributes to portfolio accumulation across households. This work connects naturally to Digonex's mission of developing rigorous, data-driven pricing algorithms grounded in behavioral responses, dynamic optimization, and robust demand modeling.

In my job-market paper, I calibrate a life-cycle heterogeneous-agent model to match wealth moments from the Survey of Consumer Finances by allowing households to differ in their portfolio returns. The model helps explain observed wealth concentration and return dispersion. In a related research project using the Health and Retirement Study, I examine how trust influences both income growth and asset returns—finding a hump-shaped relationship for returns and characterizing the distribution of the persistent component of returns to net worth. These experiences have equipped me to: (i) frame research strategies and design modeling approaches; (ii) execute econometric and simulation analysis with large micro-data sets; and (iii) translate quantitative findings into clear, rigorous written deliverables for both academic and practitioner audiences.

I am particularly excited about the opportunity to contribute to Digonex's dynamic pricing solutions, where econometric modeling, elasticity estimation, and predictive analytics drive measurable improvements in revenue and customer performance. My experience with simulation-based modeling, structural estimation, and demand-side behavioral analysis positions me well to support and enhance the algorithms that underpin Digonex's pricing products. I am also highly motivated by the chance to work on real-time pricing challenges across industries—including entertainment, hospitality, retail, and transportation—where clear model design and empirical precision are essential.

I believe I would be a strong fit at Digonex because:

1. My background in simulation-based modeling, dynamic optimization, and empirical validation aligns closely with algorithmic pricing, elasticity estimation, and revenue optimization.

2. My research on heterogeneous behavior prepares me to contribute to modeling demand responses and customer heterogeneity, which are key components of effective dynamic pricing solutions.
3. I bring strong communication skills, collaborative experience, and a commitment to producing rigorous, transparent analytic work for both technical and client-facing audiences.

I would welcome the opportunity to contribute my analytical rigor, modeling expertise, and passion for applied quantitative work to Digonex's research and pricing teams. Thank you for considering my application.

Sincerely,

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